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## Unit-IV

8. (a) Using mathematical induction, prove that 8

$$1+4+7+\dots+(3n-2)=\frac{n(3n-1)}{2}$$

- (b) Solve the recurrence relation  $a_n = 10a_{n-1} - 25a_{n-2}$   
for  $n \geq 2$  with  $a_0 = 1, a_1 = 3$  8

9. (a) Find the g.c.d. of 190 and 34. Also express the g.c.d.  
as a linear combination of the given numbers. 8

- (b) Solve the congruence  $222x \equiv 12 \pmod{18}$  8

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BCA- 2nd Semester (Full &amp; Re-appear)

Examination, May-2023

MATHEMATICAL FOUNDATION OF  
COMPUTER SCIENCE

Paper -BCA-108

Time allowed : 3 hours]

[Maximum marks : 80

**Note:** Attempt five questions in all. Question No. 1 is compulsory. Attempt four more questions by selecting one question from each unit. All questions carry equal marks.

1. (a) Give two demerits of median.
- (b) What do you mean by correlation? List various types of correlation.
- (c) Determine whether  $f(x) = x^2 + 5x + 1$  is  $O(x^2)$ .
- (d) Find  $n$ , if a complete graph having  $n$  vertices has 15 edges.
- (e) Define complete binary tree with an example.
- (f) What is merge sort?
- (g) Find the first five terms of the sequence given by recursive formula  
 $c_n = c_{n-1} + c_{n-3}, n \geq 4, c_1 = 1, c_2 = 2, c_3 = 0$
- (h) Give the statement of Division algorithm. 16

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## Unit-I

2. (a) Calculate the mean from the following data: 8

|                |      |       |       |       |       |
|----------------|------|-------|-------|-------|-------|
| Class Interval | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
| Frequency      | 15   | 20    | 35    | 20    | 10    |

- (b) Calculate the Standard Deviation for the following frequency distribution: 8

|                |       |       |       |       |       |       |        |
|----------------|-------|-------|-------|-------|-------|-------|--------|
| Class Interval | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 | 90-100 |
| Frequency      | 3     | 7     | 12    | 15    | 8     | 3     | 2      |

3. (a) Calculate Karl Pearson's coefficient of correlation from the following data: 8

|               |    |    |    |    |    |    |    |
|---------------|----|----|----|----|----|----|----|
| Husband's Age | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
| Wife's Age    | 17 | 18 | 19 | 20 | 21 | 22 | 23 |

- (b) The equations of two lines of regression are  $4x + 3y + 7 = 0$  and  $3x + 4y + 8 = 0$ .

Find:

- (i) Mean of  $x$  and  $y$   
 (ii) Regression coefficients  
 (iii) Correlation coefficient between  $x$  and  $y$

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## Unit-II

4. (a) Write an algorithm to find the roots of a quadratic equation. 8  
 (b) Write a short note on time complexity of Binary Search Algorithm. 8
5. (a) Show that the maximum number of edges in a simple graph with  $n$  vertices is  $\frac{n(n-1)}{2}$ . 8  
 (b) Differentiate between Euler graph and Hamiltonian graph with the help of examples. 8

## Unit-III

6. (a) Define spanning tree. Explain the methods for constructing a spanning tree in a simple connected graph. 8  
 (b) Prove that the number of edges is one less than the number of vertices in a tree. 8
7. (a) Convert the binary number  $(101.1101)_2$  to decimal number. 8  
 (b) Explain insertion sort technique. Use the insertion sort to sort the following list:  
 7, 8, 4, 6, 1, 0, 9 8

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